

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

1st Semester of M.Pharm [Pharmaceutics (Drug Regulatory Affairs)]

University Theory Examination December 2015

PH819.01 BASIC ANALYTICAL TECHNIQUES

Date: 16.12.2015, Wednesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 Explain principle, instrumentation and application of Differential Scanning Calorimetry (DSC). 10

Q.2 Explain principle and application of X-Ray Diffraction methods in pharmaceuticals. 10

OR

Q.2 A) Enlist methods for impurity identification. Explain application of impurity profiling 05

B) Explain principle of scanning electron microscopy. 05

Q.3 Answer the following (Any three) 15

(A) Write a short note on Reference standards.

(B) Write a principle & application of Differential Thermal Analysis (DTA).

(C) Write a note on principle of electrophoresis.

(D) Write a validation parameter for bio-analytical method.

SECTION-II

Q.4 (A) Give principle of NMR spectroscopy. How NMR spectroscopy helps in structure elucidation of compounds. 05

(B) Define the terms 05

(1) Mc-Lafferty rearrangement

(2) Metastable peak

Q.5 (A) Explain principle of UV-vis spectroscopy. 05

(B) Give principle of IR spectroscopy. What is FT-IR. 05

OR

Q.5 (A) Which detectors are used in Gas Chromatography. Explain working of any one detector. 05

(B) Write a note on principle and instrumentation of super fluid chromatography. 05

Q.6 Answer the following (Any three) 15

(A) Explain plate theory for chromatography in details.

(B) Write a brief note on Affinity chromatography.

(C) Enlist the ionization techniques used in mass spectroscopy. Discuss MALDI techniques.

(D) Write a note on HPTLC.